EIN 4343 Inventory and Supply Chain Systems

Prof. Elif Akçalı

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EIN4343 INVENTORY & SUPPLY CHAIN SYSTEMS

SUPPLY CHAIN COORDINATION: THE BULLWHIP EFFECT

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Supply chain coordination

- All stages take actions that are aligned and increase total supply chain surplus
- Requires that each stage share information and take into account the effects of its actions on the other stages
- Lack of coordination results when:
  - Objectives of different stages conflict
  - Information moving between stages is delayed or distorted

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Demand information gets distorted

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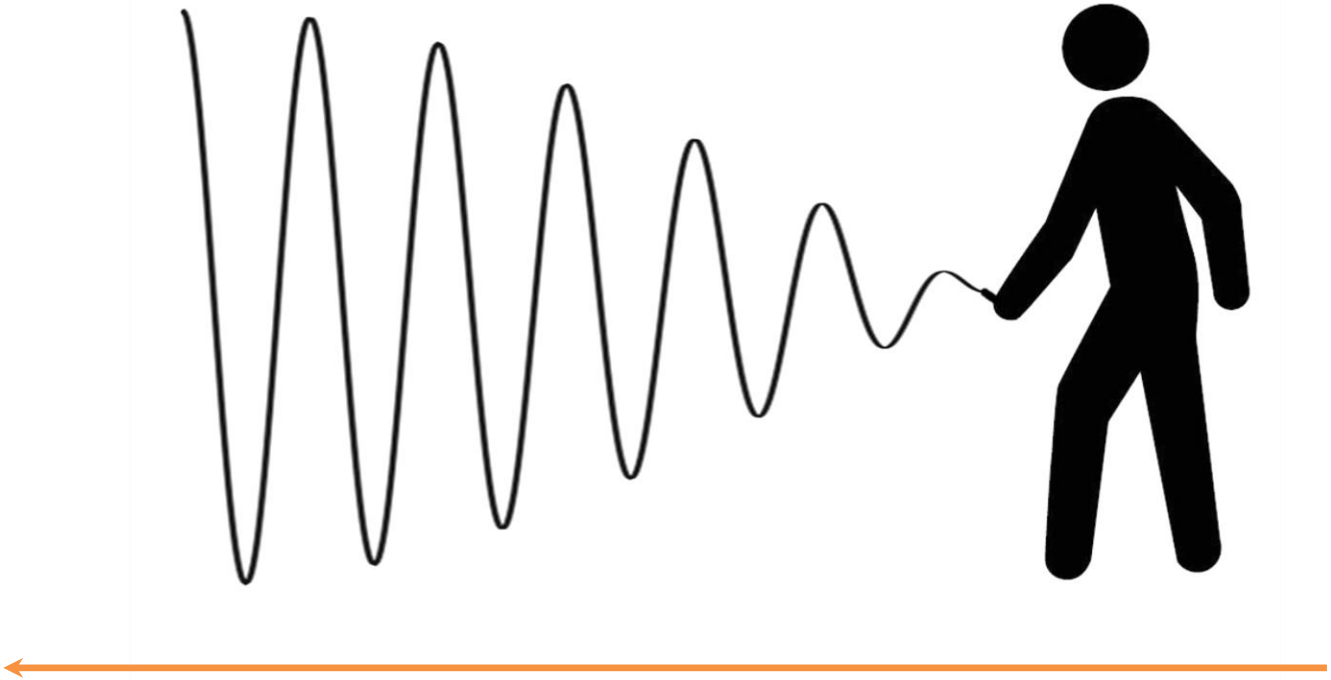
The Bullwhip Effect



SUPPLIER **MANUFACTURER** **DISTRIBUTOR** **RETAILER**

Variability in order sizes increase

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The Bullwhip Effect

- Fluctuations in orders increase as they move up the supply chain from retailers to wholesalers to manufacturers to suppliers
- Demand information gets distorted throughout the supply chain
- Due to a lack of supply chain coordination

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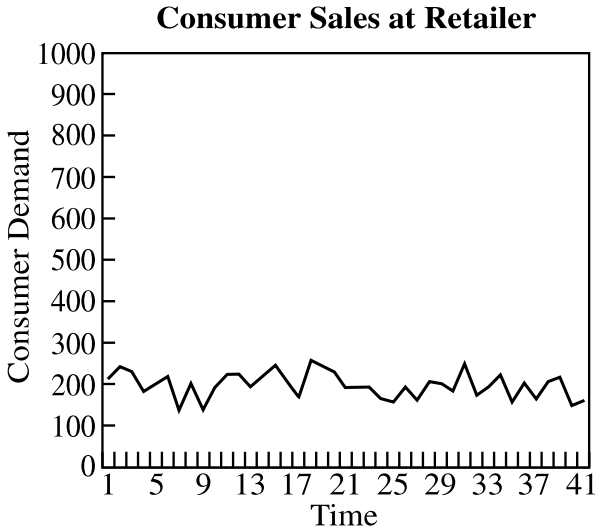
“Demand” at different stages



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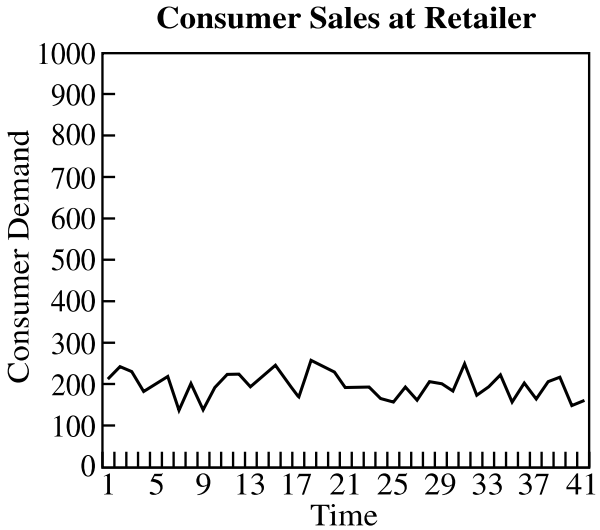

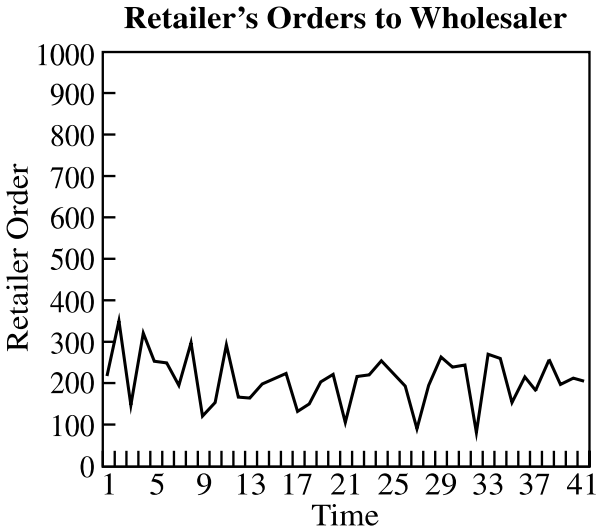

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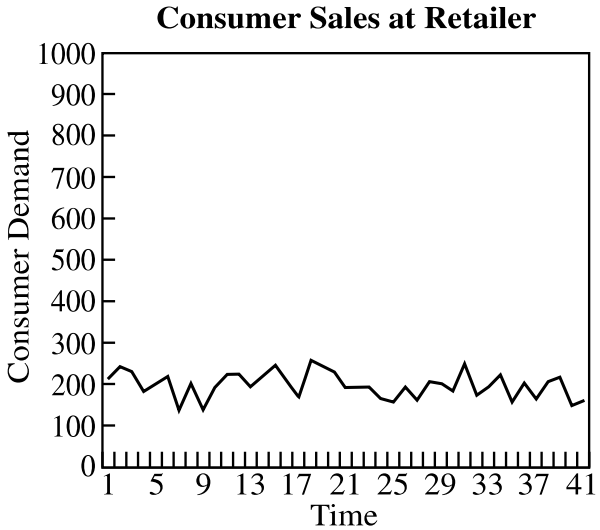

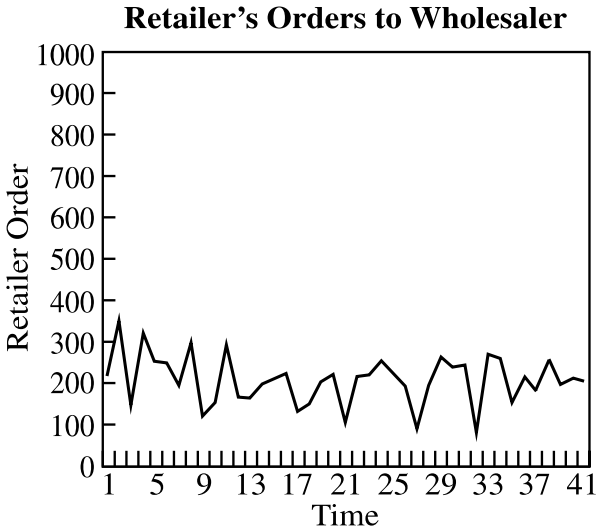

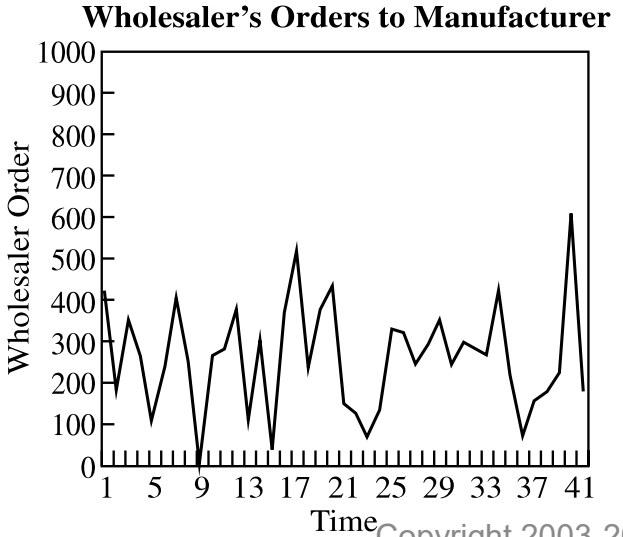

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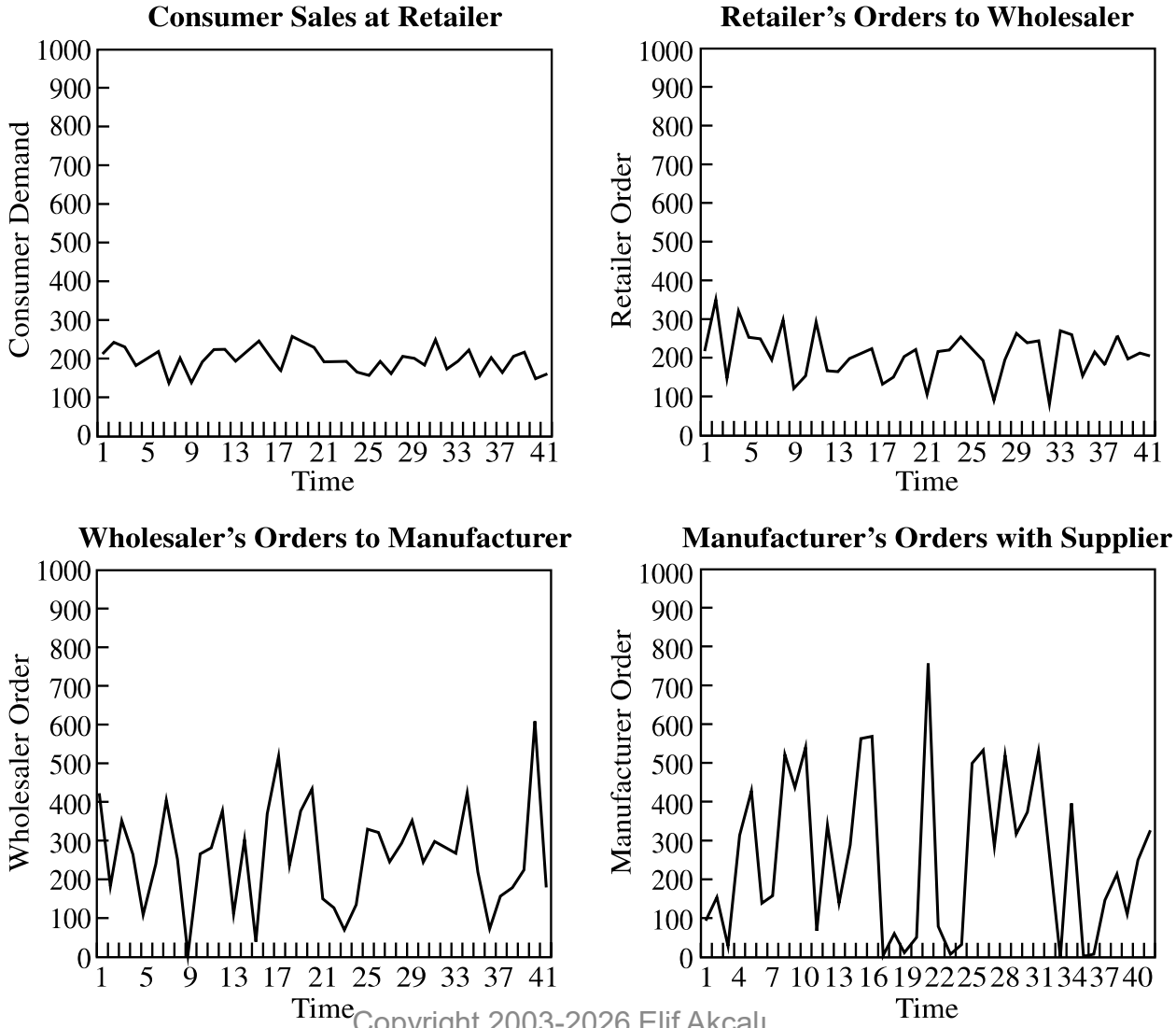
“Demand” at different stages



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Bullwhip Effect and performance

Performance Measure	Impact of the Lack of Coordination
Manufacturing cost	Increases
Inventory cost	Increases
Replenishment lead time	Increases
Transportation cost	Increases
Shipping and receiving cost	Increases
Level of product availability	Decreases
Profitability	Decreases

As a consequence of the all of the impact on the above metrics, overall profitability hurts as well

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Bullwhip Effect and performance

Trying to satisfy a demand stream that is more variable might necessitate excess capacity and use of subcontracting

Trying to satisfy a demand stream that is more variable necessitates extra inventory in the supply chain

As order sizes increase, available capacity/inventory might not be sufficient leading to longer replenishment times

As demand variability increases, transportation requirements fluctuate over time leading to an increase in cost

As transportation requirements fluctuate over time, labor requirements for shipping and receiving also increase

Fluctuation in demand/order lead to difficulties in satisfying the demand in a timely manner

As a consequence of the all of the impact on the above metrics, overall profitability hurts as well

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Obstacles to coordination

1. Incentive obstacles
2. Information processing obstacles
3. Operational obstacles
4. Pricing obstacles
5. Behavioral obstacles

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1. Incentive obstacles

- Incentives offered to different stages or participants in a supply chain lead to actions that increase variability and reduce total supply chain profits

- Local optimization within functions or stages of a supply chain

- Sales force incentives

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2. Information processing obstacles

- Demand information is distorted as it moves between different stages of the supply chain, leading to increased variability in orders within the supply chain
- Forecasting based on orders, not customer demand
- Lack of information sharing

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3. Operational obstacles

- Placing and filling orders lead to an increase in variability
- Ordering in large lots
- Large replenishment lead times
- Rationing and shortage gaming

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4. Pricing obstacles

- Pricing policies lead to an increase in variability of orders placed
- Lot-size based quantity decisions
- Price fluctuations

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5. Behavioral obstacles

- Each stage of the supply chain views its actions locally and is unable to see the impact of its actions on other stages
- Different stages of the supply chain react to the current local situation rather than trying to identify the root causes
- Different stages of the supply chain blame one another for the fluctuations
- No stage of the supply chain learns from its actions over time
- A lack of trust among supply chain partners causes them to be opportunistic at the expense of overall supply chain performance

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Managerial levers for coordination

1. Aligning goals and incentives
2. Improving information accuracy
3. Improving operational performance
4. Designing pricing strategies to stabilize orders
5. Building strategic partnerships and trust

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1. Align goals and incentives

- Align goals and incentives so that every participant in supply chain activities works to maximize total supply chain profits
- Align goals across the supply chain
- Align incentives across functions
- Pricing for coordination
- Alter sales force incentives from sell-in (to the retailer) to sell-through (by the retailer)

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2. Improve information accuracy

- Sharing point of sale data
- Implementing collaborative forecasting and planning
- Designing single-stage control of replenishment
  - Continuous replenishment programs (CRP)
  - Vendor managed inventory (VMI)

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3. Improve operational performance

- Reducing replenishment lead time
- Reducing lot sizes
- Rationing based on past sales and sharing information to limit gaming

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4. Stabilize orders through pricing

- Encouraging retailers to order in smaller lots and reduce forward buying
- Moving from lot size-based to volume-based quantity discounts
- Stabilizing pricing
- Building strategic partnerships and trust

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5. Build strategic partnerships and trust

- Improving relationships to reduce transaction cost (e.g., reduce duplicated effort, focus on strength)

- Examples
  - Continuous replenishment programs
  - Vendor managed inventory
  - Collaborative planning, forecasting and replenishment

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Achieving coordination in practice

1. Quantify the Bullwhip Effect
2. Get top management commitment for coordination and devote resources to coordination
3. Focus on communication with other stages
4. Try to achieve coordination in the entire supply chain network
5. Use technology to improve connectivity in the supply chain
6. Share the benefits of coordination equitably

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SUPPLY CHAIN COORDINATION: A SIMPLE MODEL

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A two-stage system

- A supplier-retailer pair

- Retailer
  - Faces constant demand rate λ from customers
  - Incurs a fixed order cost of K to place an order from the supplier and a purchasing cost of c/unit
  - Also incurs annual inventory holding cost of h/unit/year

- Supplier
  - Incurs a fixed order cost of βK to place an order from the supplier Also incurs annual inventory holding cost of αh/unit/year
  - Note that $\beta > 1$ and $0 < \alpha < 1$

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Supplier driven channel

- Supplier has the channel power
  - Makes the replenishment quantity decision

$$q_S^* = \sqrt{\frac{2\beta K\lambda}{\alpha h}}$$

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Retailer driven channel

- Retailer has the channel power
  - Makes the replenishment quantity decision

$$q_R^* = \sqrt{\frac{2K\lambda}{h}}$$

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Integrated channel

- Supplier-retailer pair make the replenishment quantity decision together

$$q_I^* = \sqrt{\frac{2(1+\beta)K\lambda}{(1+\alpha)h}}$$

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Example 8.1: Sony & Best Buy

Sony markets its televisions through Best Buy stores all around nation as well as other “brick-and-mortar” and “on-line” channels.

Suppose that the demand for a particular television type at a particular Best Buy retail location can assumed to be constant. Therefore, both Sony and the Best Buy store use economic order quantity approach to determine the replenishment quantity.

Suppose that the unit cost for the television set is $1,200. The fixed order cost for Sony is $12,000, whereas it is $250 for Best Buy is $250. The annual inventory holding rate is 25% and 40% for Sony and Best Buy, respectively. Suppose that the annual demand for the television is 450 units per year.

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Example 8.1: Sony & Best Buy

1. If Sony has the channel power, i.e., Sony determines the replenishment quantity using its fixed ordering cost and inventory holding rate, calculate the corresponding economic order quantity. Calculate the inventory ordering and holding costs incurred by Sony and Best Buy.

$$q_S^* = \sqrt{\frac{2K_S\lambda}{i_S c}} = \sqrt{\frac{(2)(12{,}000)(450)}{(0.25)(1{,}200)}} \quad = 189.74 \text{ units}$$

$$C_S(q_S^*) = \frac{K_S\lambda}{q_S^*} + \frac{i_S c q_S^*}{2} = \frac{(12{,}000)(450)}{189.74} + \frac{(0.25)(1{,}200)(189.74)}{2}$$

$$C_{BB}(q_S^*) = \frac{K_{BB}\lambda}{q_S^*} + \frac{i_{BB} c q_S^*}{2} = \frac{(250)(450)}{189.74} + \frac{(0.40)(1{,}200)(189.74)}{2}$$

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Example 8.1: Sony & Best Buy

2. If Best Buy has the channel power, i.e., Best Buy determines the replenishment quantity using its fixed ordering cost and inventory holding rate, calculate the corresponding economic order quantity. Calculate the inventory ordering and holding costs incurred by Sony and Best Buy.

$$q_{BB}^* = \sqrt{\frac{2K_{BB}\lambda}{i_{BB}c}} = \sqrt{\frac{(2)(250)(450)}{(0.40)(1{,}200)}} = 21.65 \text{ units}$$

$$C_S(q_{BB}^*) = \frac{K_S\lambda}{q_{BB}^*} + \frac{i_S c q_{BB}^*}{2} = \frac{(12{,}000)(450)}{21.65} + \frac{(0.25)(1{,}200)(21.65)}{2}$$

$$C_{BB}(q_{BB}^*) = \frac{K_{BB}\lambda}{q_{BB}^*} + \frac{i_{BB} c q_{BB}^*}{2} = \frac{(250)(450)}{21.65} + \frac{(0.40)(1{,}200)(21.65)}{2}$$

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Example 8.1: Sony & Best Buy

3. 3. If Sony and Best Buy were to determine an ordering quantity together. Determine the corresponding economic order quantity and calculate the total cost incurred by Sony and Best Buy.

$$q^*_{S+BB} = \sqrt{\frac{2(K_S + K_{BB})\lambda}{(i_S + i_{BB})c}} = \sqrt{\frac{(2)(12{,}000 + 250)(450)}{(0.25 + 0.40)(1{,}200)}} = 118.89 \text{ units}$$

$$C_S(q^*_{S+BB}) = \frac{K_S\lambda}{q^*_{S+BB}} + \frac{i_S c q^*_{S+BB}}{2} = \frac{(12{,}000)(450)}{118.89} + \frac{(0.25)(1{,}200)(118.89)}{2}$$

$$C_{BB}(q^*_{S+BB}) = \frac{K_{BB}\lambda}{q^*_{S+BB}} + \frac{i_{BB} c q^*_{S+BB}}{2} = \frac{(250)(450)}{118.89} + \frac{(0.40)(1{,}200)(118.89)}{2}$$

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Example 8.1: Discussion

Leader	Order Quantity (units)	Sony's Cost ($/year)	Best Buy's Cost ($/year)	Channel Cost ($/year)
Sony	189.74	$56,921	$46,130	$103,051
Best Buy	21.65	$252,663	$10,392	$263,055
Integrated	118.89	$63,254	$29,480	$92,734

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